Phase Validation

CHECKLIST

✓	Criterion	What to verify	Pts
1. CONTAINERIZATION — 4 pts			**4 pts**
x	**Separate Dockerfiles**	One Dockerfile per service: Frontend, Backend, DB	**2.5**
x	**Multi-stage build**	Build stage separated from runtime stage	
x	**No build tools in prod image**	npm, compilers, etc. absent from final image	
x	**Security hardening**	Non-root user configured + .dockerignore present	
x	**Docker network configured**	Services on the same network	**1.5**
x	**Frontend → Backend → DB chain**	Full communication chain works, no runtime errors	
2. KUBERNETES ORCHESTRATION — 6 pts			**6 pts**
x	**Valid YAML manifests**	Deployment, Service, ConfigMap, Secret all present	**2**
x	**Master node healthy**	kube-apiserver, etcd, kube-scheduler, kube-controller-manager all running	
x	**Worker node healthy**	kubelet, kube-proxy, container runtime all running	
x	**Pods in Running state**	No CrashLoopBackOff anywhere (kubectl get pods -A)	**1.5**
x	**Service externally accessible**	NodePort or LoadBalancer configured and reachable	
x	**ConfigMap used**	Non-sensitive config stored in ConfigMap (not hardcoded)	**1**
x	**Secrets used**	Passwords/tokens stored in Secrets (not in code)	
x	**Replicas configured**	At least 2 replicas on key deployments	**1.5**
x	**HPA enabled**	Horizontal Pod Autoscaler configured (bonus: demo live)	
3. INFRASTRUCTURE & AUTOMATION (IaC) — 6 pts			**6 pts**
x	**OpenStack cluster accessible**	Instances up with correct CPU/RAM/storage	**1.5**
x	**Network configured**	VPC/subnet, security groups/firewall rules in place	

x	**Ansible Level 1 (min)**	Separate playbooks for K8s + Monitoring stack	**1.5**
x	**Ansible Level 2 (better)**	Structured Ansible roles: K8s role + Monitoring role	**2.5**
x	**Ansible Level 3 (max)**	Heat template → auto VM provisioning → auto Ansible, zero manual steps	**3**
x	**App accessible**	URL or IP reachable, frontend + backend functional	**1.5**
x	**Deployment reproducible**	Someone else can redeploy from scratch using your doc	
x	**Steps clearly documented**	README with all commands, variables, and verification steps	
4. MONITORING & AI OPS — 4 pts			**4 pts**
x	**Prometheus + Grafana deployed**	Stack operational, not just installed	**1**
x	**Coverage: nodes AND pods**	Dashboards show both node metrics and pod metrics	
x	**Dashboards readable**	Panels labeled, units correct, relevant metrics shown	
x	**Alert rules configured**	At least 2 distinct alert rules set up	**1**
x	**2 alerts triggered live**	Both alerts fire during the demo (not just configured)	
x	**AI scenario chosen & ready**	One scenario prepared	**1**
☐	**Demo is clear and structured**	Well-paced, logical order, team confident on questions	**1**

PART 2 — SCREENSHOT

	What to capture	content
KUBERNETES		
📷	**kubectl get nodes**	All nodes in Ready state

```
ubuntu@k8s-master:~$ kubectl get nodes
NAME          STATUS   ROLES           AGE   VERSION
k8s-master    Ready    control-plane   35h   v1.31.14
k8s-worker1   Ready    <none>          35h   v1.31.14
k8s-worker2   Ready    <none>          35h   v1.31.14
```

📷	**kubectl get pods -A**	All pods in Running state, 0 CrashLoopBackOff —

```
ubuntu@k8s-master:~$ kubectl get pods -A
NAMESPACE      NAME                                                READY   STATUS    RESTARTS          AGE
app            backend-6d665b586c-l5n4q                            1/1     Running   4 (3h14m ago)     5h26m
app            backend-6d665b586c-wqm8p                            1/1     Running   1 (3h14m ago)     5h26m
app            frontend-797894b57c-ckd84                           1/1     Running   1 (3h15m ago)     5h24m
app            frontend-797894b57c-tn755                           1/1     Running   1 (3h14m ago)     5h24m
app            mysql-c584d746c-fwtkz                               1/1     Running   1 (3h14m ago)     5h30m
default        sre-anomaly-agent-594b57747d-bvq2b                  1/1     Running   0                 3h11m
default        sre-anomaly-agent-594b57747d-czm82                  1/1     Running   0                 3h11m
kube-flannel   kube-flannel-ds-47pvt                               1/1     Running   3 (3h16m ago)     6h56m
kube-flannel   kube-flannel-ds-j6dd2                               1/1     Running   1 (3h15m ago)     6h56m
kube-flannel   kube-flannel-ds-w8nmc                               1/1     Running   1 (3h14m ago)     6h56m
kube-system    coredns-765dd469d6-f2kqb                            1/1     Running   1 (3h14m ago)     5h39m
kube-system    coredns-765dd469d6-jpq7t                            1/1     Running   1 (3h15m ago)     5h39m
kube-system    csi-cinder-controllerplugin-6ccc56cd9f-wqpqq        6/6     Running   24 (169m ago)     9h
kube-system    csi-cinder-nodeplugin-4cgc4                         3/3     Running   16 (3h14m ago)    32h
kube-system    csi-cinder-nodeplugin-9mlpq                         3/3     Running   22 (3h17m ago)    32h
kube-system    csi-cinder-nodeplugin-vkdb5                         3/3     Running   19 (3h15m ago)    32h
kube-system    etcd-k8s-master                                     1/1     Running   32 (3h17m ago)    35h
kube-system    kube-apiserver-k8s-master                           1/1     Running   15 (3h17m ago)    35h
kube-system    kube-controller-manager-k8s-master                  1/1     Running   9 (3h17m ago)     35h
kube-system    kube-proxy-2kvgs                                    1/1     Running   1 (3h15m ago)     5h43m
kube-system    kube-proxy-656l6                                    1/1     Running   1 (3h17m ago)     5h43m
kube-system    kube-proxy-v9l78                                    1/1     Running   1 (3h14m ago)     5h43m
kube-system    kube-scheduler-k8s-master                           1/1     Running   9 (3h17m ago)     35h
kube-system    metrics-server-cf745b4cc-s9px6                      1/1     Running   0                 139m
kube-system    node-exporter-dkdd9                                 1/1     Running   0                 134m
kube-system    node-exporter-gqb8n                                 1/1     Running   0                 134m
kube-system    node-exporter-kg59s                                 0/1     Pending   0                 82m
kube-system    openstack-cloud-controller-manager-5swjb            1/1     Running   12 (3h16m ago)    14h
monitoring     alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2   Running   0           100m
monitoring     monitoring-grafana-7876ccdbc5-nwl6q                 3/3     Running   0                 100m
```

📷	**kubectl get svc -A**	Services with their IPs/ports

```
ubuntu@k8s-master:~$ kubectl get svc -A
NAMESPACE     NAME                                             TYPE           CLUSTER-IP       EXTERNAL-IP     PORT(S)                        AGE
app           backend                                          ClusterIP      10.103.160.111   <none>          8087/TCP                       5h31m
app           frontend                                         LoadBalancer   10.105.104.211   192.168.1.169   80:30980/TCP                   5h25m
app           mysql                                            ClusterIP      None             <none>          3306/TCP                       5h31m
default       kubernetes                                       ClusterIP      10.96.0.1        <none>          443/TCP                        35h
default       sre-anomaly-agent                                ClusterIP      10.107.145.1     <none>          80/TCP                         3h12m
kube-system   kube-dns                                         ClusterIP      10.96.0.10       <none>          53/UDP,53/TCP,9153/TCP         35h
kube-system   metrics-server                                   ClusterIP      10.101.69.22     <none>          443/TCP                        149m
kube-system   monitoring-kube-prometheus-coredns               ClusterIP      None             <none>          9153/TCP                       110m
kube-system   monitoring-kube-prometheus-kube-controller-manager   ClusterIP  None             <none>          10257/TCP                      110m
kube-system   monitoring-kube-prometheus-kube-etcd             ClusterIP      None             <none>          2381/TCP                       110m
kube-system   monitoring-kube-prometheus-kube-proxy            ClusterIP      None             <none>          10249/TCP                      110m
kube-system   monitoring-kube-prometheus-kube-scheduler        ClusterIP      None             <none>          10259/TCP                      110m
kube-system   monitoring-kube-prometheus-kubelet               ClusterIP      None             <none>          10250/TCP,4194/TCP,10255/TCP   109m
kube-system   node-exporter                                    ClusterIP      10.110.146.237   <none>          9100/TCP                       115m
monitoring    alertmanager-operated                            ClusterIP      None             <none>          9093/TCP,9094/TCP,9094/UDP     109m
monitoring    monitoring-grafana                               ClusterIP      10.106.176.246   <none>          80/TCP                         110m
monitoring    monitoring-kube-prometheus-alertmanager          ClusterIP      10.106.150.135   <none>          9093/TCP,8080/TCP              110m
monitoring    monitoring-kube-prometheus-operator              ClusterIP      10.102.189.168   <none>          443/TCP                        110m
monitoring    monitoring-kube-prometheus-prometheus            NodePort       10.103.177.89    <none>          9090:30151/TCP,8080:32710/TCP  110m
monitoring    monitoring-kube-state-metrics                    ClusterIP      10.105.92.169    <none>          8080/TCP                       110m
monitoring    monitoring-prometheus-node-exporter              ClusterIP      10.109.92.170    <none>          9100/TCP                       110m
monitoring    prometheus-operated                              ClusterIP      None             <none>          9090/TCP                       109m
ubuntu@k8s-master:~$
```

📷	**kubectl get configmap**	ConfigMaps listed in the cluster

```
ubuntu@k8s-master:~$ kubectl get configmap -n webapp
NAME               DATA   AGE
backend-config     5      91s
frontend-config    1      78s
kube-root-ca.crt   1      6h49m
mysql-config       1      103s
ubuntu@k8s-master:~$
```

📷	**kubectl get secret**	Secrets listed in the cluster

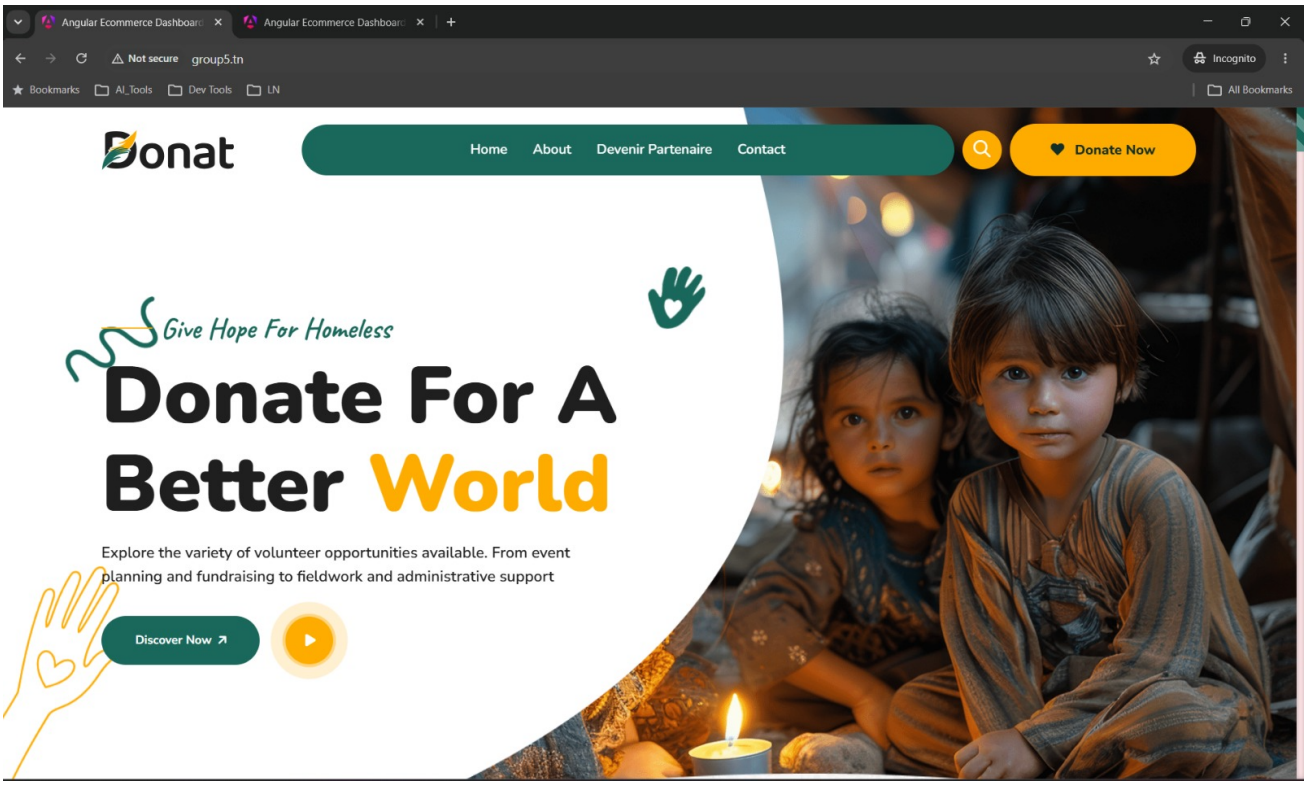
```
ubuntu@k8s-master:~$ kubectl get secret -n webapp
NAME             TYPE     DATA   AGE
backend-secret   Opaque   3      20s
mysql-secret     Opaque   3      27s
ubuntu@k8s-master:~$
```

📷	**kubectl get hpa**	HPA configured and showing current/target replicas

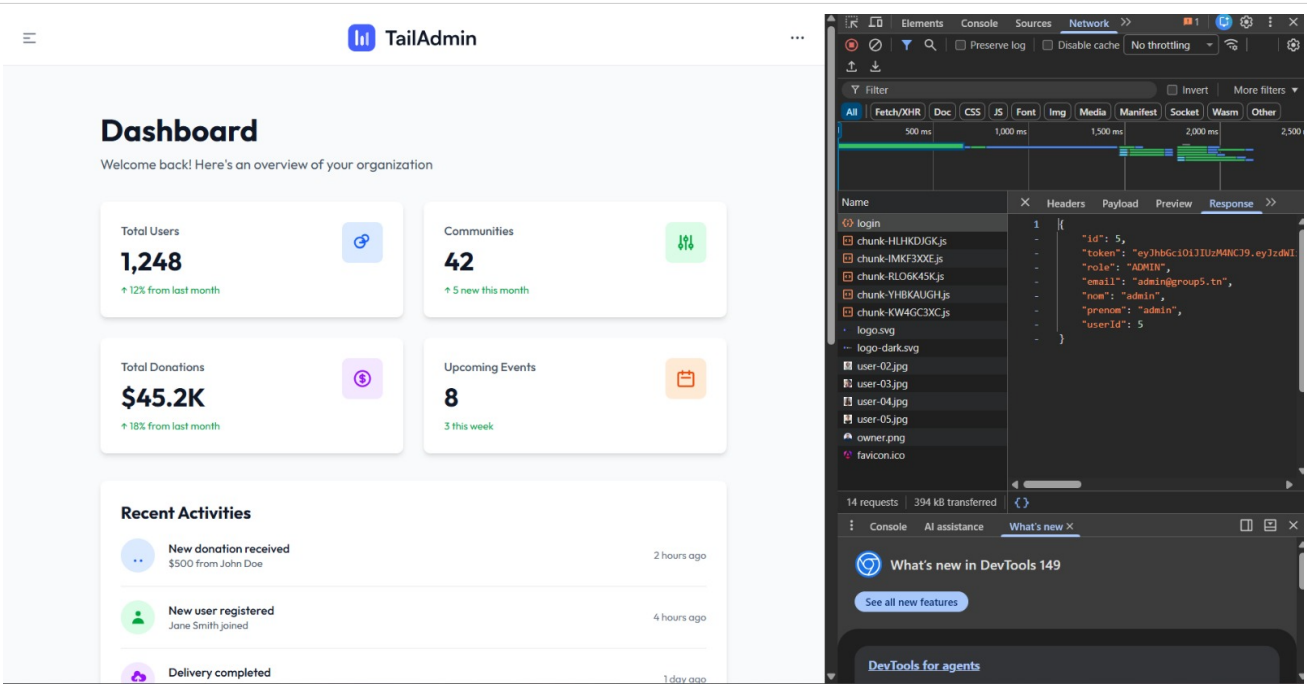
```
ubuntu@k8s-master:~$ kubectl get hpa -n app
NAME           REFERENCE              TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
backend-hpa    Deployment/backend     cpu: <unknown>/70%   2         5         2          5h
frontend-hpa   Deployment/frontend    cpu: <unknown>/70%   2         4         2          5h
ubuntu@k8s-master:~$
```

APPLICATION

📷	**Frontend in browser**	App fully loaded at the public IP/URL



📷	**Backend responding**	API endpoint returning data (browser or curl)



📷	**DB connection working**	A DB query or backend log confirming DB connection

```
ubuntu@k8s-master:~$ kubectl exec -it mysql-c584d746c-fwtkz -n app -- bash
bash-5.1# mysql -u appuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 175
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use appdb ;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select * from utilisateur ;
+----+-------------------------+-----------------+------------------+-------------+-------------------------------------------------------------+--------+---------+--------+-----------------+-----------------------+---------+-----------+-----------+-----------+
| id | adresse                 | association_rib | date_inscription | disponibilite | document_verification | domaine_activite | email                 | est_verifiee | mot_de_passe                                                | nom     | photo | prenom  | role        | statut  | statut_verification | telephone | vehicule  | zone_couverture |
+----+-------------------------+-----------------+------------------+-------------+-------------------------------------------------------------+--------+---------+--------+-----------------+-----------------------+---------+-----------+-----------+-----------+
|  1 | rue eur                 | NULL            | 2026-06-26       | 0x00        | NULL                  |                1 | oussemamrabet110@gmail.com | NULL         | $2a$10$oCWj.NdQxDdvjQOOpUFC7.TXae3n4cDY1ZsEfeZRR1.P/QRJVnw8i | mrabet  | NULL  | oussama | DONOR       | ENABLED |                   1 | 55814108  | NULL      | NULL            |
|  2 | rue essaha, ariana      | NULL            | 2026-06-26       | 0x00        | NULL                  |                1 | oussemamrab@gmail.com      | NULL         | $2a$10$bNRofsyypJxwgeVoapnXWO18wwBZl1x1jSzefWnEEkLOa14FUhTSO | mrabet  | NULL  | oussama | DONOR       | ENABLED |                   1 | 55814108  | NULL      | NULL            |
|  3 | 28 rue 4137 Somrane Tunis | NULL          | 2026-06-26       | 0x00        | NULL                  |                1 | islem24762048@gmail.com    | NULL         | $2a$10$fgCda74siVI0BMlzeWCmvuT\VHfezCJEyIUaARugWXbRVdgZ04b2K | Ghariani | NULL  | Islem   | DONOR       | ENABLED |                   1 | 55092584  | NULL      | NULL            |
|  5 | admin                   | NULL            | 2026-06-26       | NULL        | NULL                  |                1 | admin@group5.tn            | NULL         | $2a$10$bNRofsyypJxwgeVoapnXWO18wwBZl1x1jSzefWnEEkLOa14FUhTSO | admin   | NULL  | admin   | ADMIN       | ENABLED |                   1 | NULL      | NULL      | NULL            |
|  6 | 28 rue 4137 Somrane Tunis | NULL          | 2026-06-26       | 0x00        | NULL                  |                1 | islemghariani.dev@gmail.com | NULL        | $2a$10$R1wWlFTDUTYhwlxmhVAVQu69insMljD/D6nCy4jeyXcNDTy1zkBkm | Ghariani | NULL  | Islem   | DONOR       | ENABLED |                   1 | 55092584  | NULL      | NULL            |
|  7 | 28 rue 4137 Somrane Tunis | NULL          | 2026-06-27       | 0x00        | NULL                  |                1 | islemghariaooont.dev@gmail.com | NULL      | $2a$10$wF/wIaHl38pILDUUr5Rj9.sV3sOe.yTjGHIakYKKNjREDz0RuOjUS | Ghariani | NULL  | Islem   | DONOR       | ENABLED |                   1 | 55092584  | NULL      | NULL            |
|  8 | KAYO GHAZELA            | NULL            | 2026-06-27       | 0x00        | NULL                  |                1 | zainebkorbi1234@gmail.com  | NULL         | $2a$10$HvQoOWYchRgrRbVEfLIwFelUnHzcQVR8D9Dw6yD1H1GbI.TYR6J1G | Zaineb  | NULL  | Korbi   | DONOR       | ENABLED |                   1 | 26548920  | NULL      | NULL            |
|  9 | rue kayo                | NULL            | 2026-06-27       | 0x00        | NULL                  |                1 | zainebkorbi11234@gmail.com | NULL         | $2a$10$.nG5YImI9gZobFd2bGiSvusFeNoR5oJ9WZrMvsB5RgVA0/f3xBAky | zitouna | NULL  | Korbi   | DELIVERY_PERSON | ENABLED |                   1 | 52478963  | GRAND I10 | TUNIS           |
+----+-------------------------+-----------------+------------------+-------------+-------------------------------------------------------------+--------+---------+--------+-----------------+-----------------------+---------+-----------+-----------+-----------+
8 rows in set (0.00 sec)

mysql>
```

ANSIBLE / IAC

📷	**Ansible PLAY RECAP**	Final recap showing 0 failed, all hosts green

Live

Live

Live

📷	**OpenStack instances**	VMs listed in OpenStack dashboard with status

Instances

Instance ID = ▾ | Filter | Launch Instance | Delete Instances | More Actions ▾

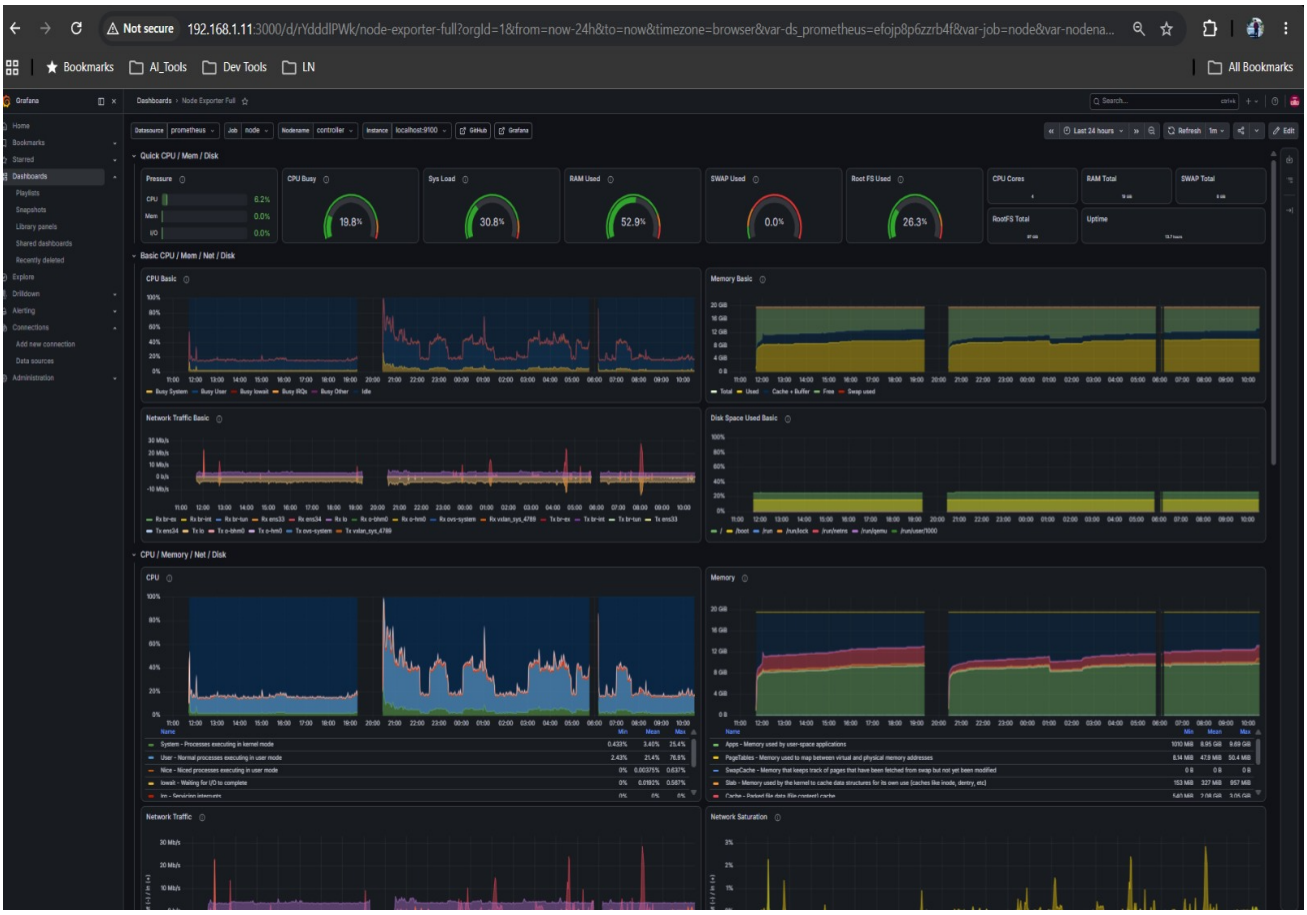
Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	k8s-worker1	ubuntu-22.04	10.10.0.8, 192.168.1.102	m1.medium	my-key	Active	nova	None	Running	3 days, 8 hours	Create Snapshot ▾
☐	k8s-master	ubuntu-22.04	10.10.0.169, 192.168.1.204	m1.medium	my-key	Active	nova	None	Running	3 days, 8 hours	Create Snapshot ▾
☐	k8s-worker2	ubuntu-22.04	10.10.0.38, 192.168.1.231	m1.medium	my-key	Active	nova	None	Running	3 days, 8 hours	Create Snapshot ▾

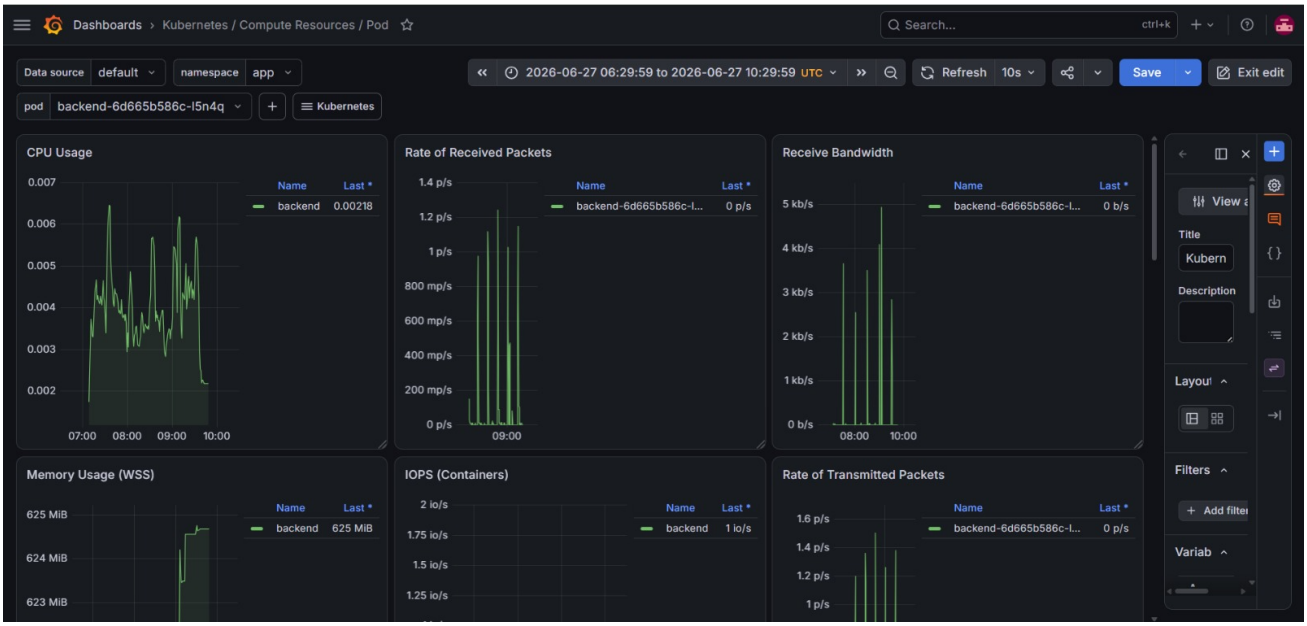
Displaying 3 items

MONITORING

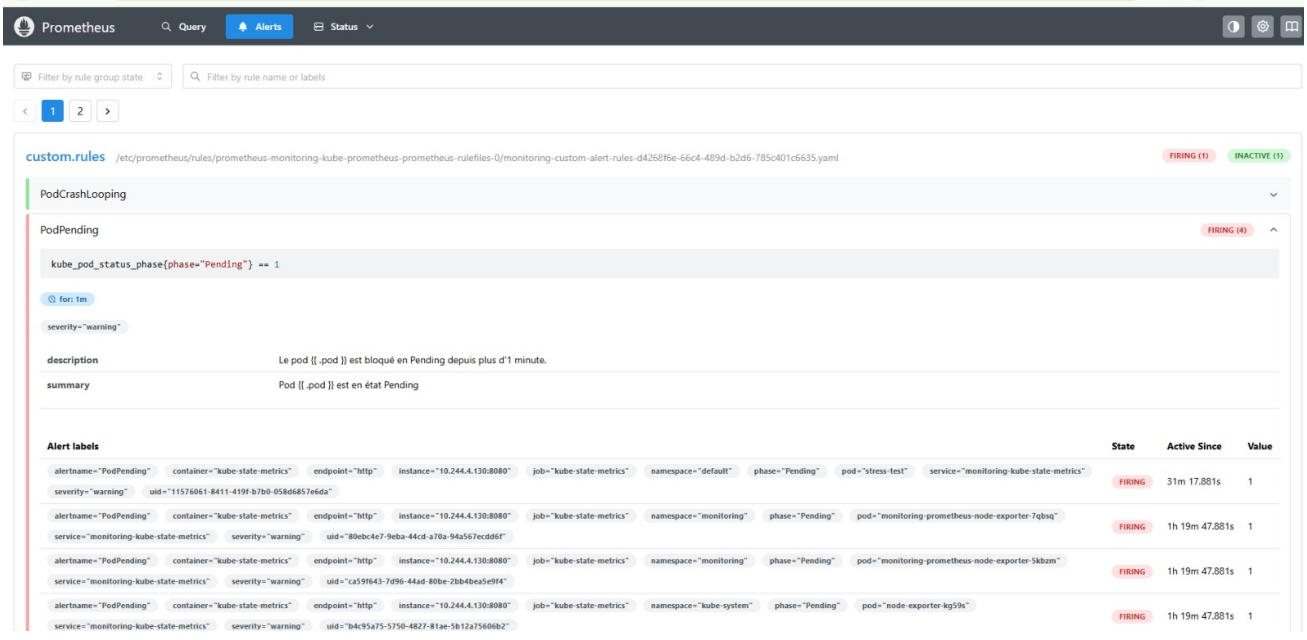
📷	**Grafana node dashboard**	CPU, RAM, disk panels for cluster nodes



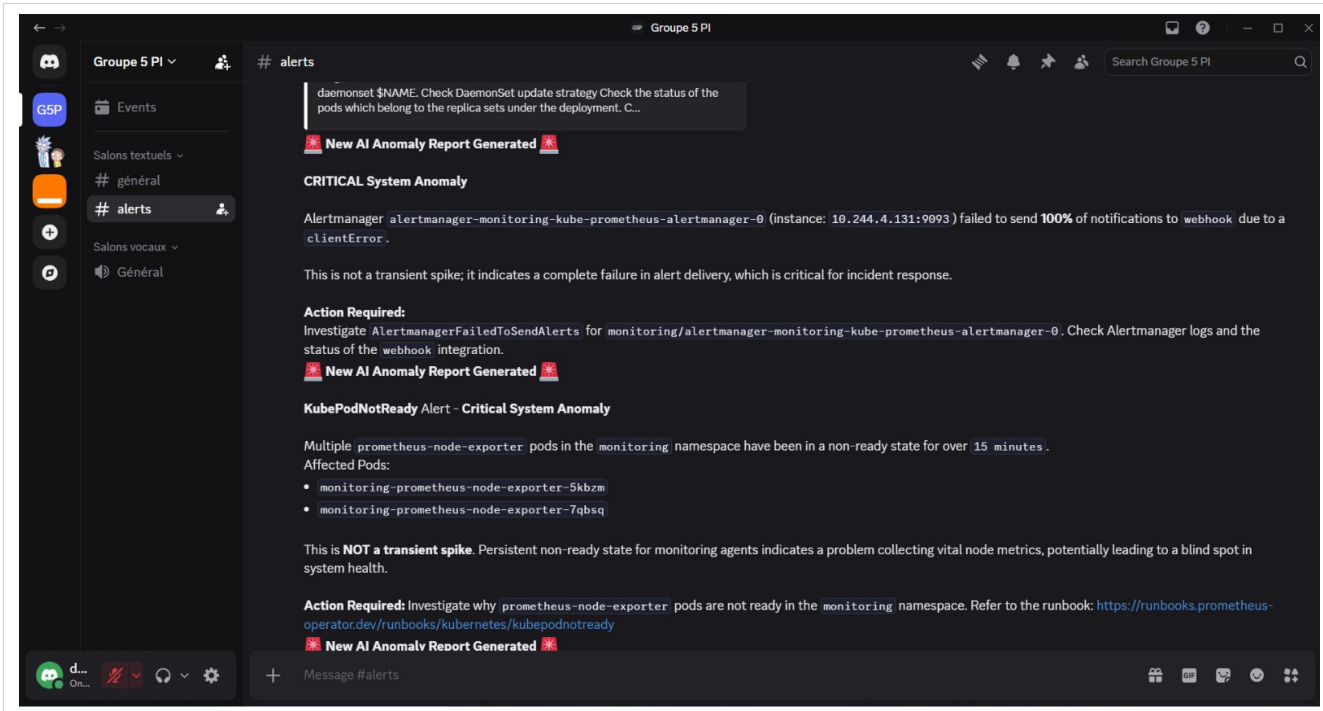
📷	**Grafana pod dashboard**	Pod-level metrics (restarts, CPU, memory)



	Alert Manager of pod pending and loopbackoff)	Alert visibly triggered in Prometheus



	AI scenario in action


Groupe 5 PI
Search Groupe 5 PI
Groupe 5 PI
Events
Salons textuels
général
alerts
Salons vocaux
Général
alerts
daemonset $NAME. Check DaemonSet update strategy Check the status of the pods which belong to the replica sets under the deployment. C...
New AI Anomaly Report Generated
CRITICAL System Anomaly
Alertmanager alertmanager-monitoring-kube-prometheus-alertmanager-0 (instance: 10.244.4.131:9093) failed to send 100% of notifications to webhook due to a clientError.
This is not a transient spike; it indicates a complete failure in alert delivery, which is critical for incident response.
Action Required:
Investigate AlertmanagerFailedToSendAlerts for monitoring/alertmanager-monitoring-kube-prometheus-alertmanager-0. Check Alertmanager logs and the status of the webhook integration.
New AI Anomaly Report Generated
KubePodNotReady Alert - Critical System Anomaly
Multiple prometheus-node-exporter pods in the monitoring namespace have been in a non-ready state for over 15 minutes.
Affected Pods:
monitoring-prometheus-node-exporter-5kbzm
monitoring-prometheus-node-exporter-7qbsq
This is NOT a transient spike. Persistent non-ready state for monitoring agents indicates a problem collecting vital node metrics, potentially leading to a blind spot in system health.
Action Required: Investigate why prometheus-node-exporter pods are not ready in the monitoring namespace. Refer to the runbook: https://runbooks.prometheus-operator.dev/runbooks/kubernetes/kubepodnotready
New AI Anomaly Report Generated
Message #alerts